Missing Value Handling &
Custom Categorical Encoding Technique

1. Dataset Description

The Diabetes dataset has 199 records and 9 attributes of mixed type. Categorical attributes are integer-encoded so that all techniques operate on a uniform numeric matrix, and the mappings are stored to decode the results.

Attribute	Type
age, bmi, HbA1c_level, blood_glucose_level	numeric
gender, smoking_history	categorical
hypertension, heart_disease	binary (0/1)
diabetes	binary classification

136 of the 199 records are fully complete and 63 contain at least one missing value. Eight columns require imputation: gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level and blood_glucose_level.

2. Custom Encoding Technique: Ordinal-Frequency Statistical Encoding

Ordinal-Frequency Statistical Encoding is a hybrid encoding technique that combines information from both Ordinal Encoding and Frequency Encoding to map each categorical value to a single numeric value. It does not use the target variable, so there is no risk of data leakage; the encoding is based entirely on the feature's own distribution.

2.1 Definition

For each categorical column, the distinct categories are sorted and given an ordinal rank (0, 1, 2, …). Each category's frequency proportion its share of the observed (non-missing) values is then added to that rank:

encoded(category) = ordinal_rank(category) + frequency_proportion(category)

This formulation ensures that each category remains uniquely identifiable while simultaneously embedding information about its prevalence in the dataset.

2.2 Encoded values on this dataset

Applying OFSE to the two categorical columns produces the following codes:

Column	Category	Ordinal rank	Frequency	Encoded value
gender	Female	0	0.623	0.623
gender	Male	1	0.377	1.377

Column	Category	Ordinal rank	Frequency	Encoded value
smoking_history	No Info	0	0.324	0.324
smoking_history	current	1	0.074	1.074
smoking_history	ever	2	0.043	2.043
smoking_history	former	3	0.106	3.106
smoking_history	never	4	0.404	4.404
smoking_history	not current	5	0.048	5.048

2.3 Decoding

After imputation, a predicted numeric value is mapped back to a category by nearest-match decoding: the encoded value closest to the prediction is found and its category returned. For example, an imputed value of 4.30 in smoking_history is closest to 4.404 and decodes to “never”.

2.4 Rationale for Combining Ordinal and Frequency Information

a. Ordinal ranks ensure separation between categories.
b. Each categorical feature is reduced to a single numeric column, avoiding the high dimensionality of one-hot encoding.

2.5 Limitations

a. The ordinal part orders nominal categories (alphabetically by default). This order is arbitrary for variables with no natural ranking, though the frequency offset and the flexibility of tree models reduce its impact.
b. Nearest-value decoding may be sensitive to floating-point precision and prediction noise.

3. Methodology

The approach turns imputation into a set of per-column prediction tasks and selects the best predictor for each column from held-out performance.

a. Split the data into complete records and records containing missing values.
b. Divide the complete records into 80% training and 20% testing.
c. For each column that has missing values, treat that column as the target and all other columns as features; train all fifteen techniques and measure MSE on the held-out 20%.
d. The technique with the lowest MSE is chosen as the best method for that column, different columns may select different techniques.
e. Each column's winning technique is retrained on all complete records and used to fill that column's real gaps. Because feature columns may themselves be missing, a temporary mean-filled scaffold supplies the features, and only the

originally-missing cells are overwritten; binary columns are rounded and categoricals decoded to their labels.

4. Techniques Compared

The candidates span simple baselines, linear models, tree ensembles, boosting, instance-based and kernel methods:

Group	Techniques
Baselines	Mean, Median, Mode
Linear models	Linear Regression, Ridge, Lasso, Bayesian Ridge
Decision Tree	Decision Tree, Random Forest, Extra Trees
Boosting	Gradient Boosting, HistGradient Boosting, AdaBoost
Instance	K-Nearest-Neighbours, Support Vector Regression

Distance- and scale-sensitive methods (KNN, SVR, linear models) are standardized inside a pipeline, while tree-based methods use the raw encoded matrix.

5. Results

5.1 Best technique selected for each column

Each column with missing values, the technique chosen for it, its held-out MSE, and the error reduction versus the Mean baseline:

Column	Best technique	MSE
age	KNN	412.734
bmi	SVR	32.666
HbA1c_level	Linear Regression	0.942
blood_glucose_level	Linear Regression	2061.342
smoking_history	Random Forest	3.261
heart_disease	Ridge	0.065
gender	Mean	0.127
hypertension	Mean	0.048

6. Discussion and Key Findings

a. **The best technique is column-specific.**

The results show that no single technique consistently performs best across all variables. Different columns selected different winning methods, including Linear

Regression, KNN, SVR, Random Forest, Ridge, and Mean imputation. This supports the use of a per-column selection strategy rather than applying a single imputation method to the entire dataset.

b. **Numeric variables benefit most from predictive imputation.**

Columns such as HbA1c_level, blood_glucose_level, age, and bmi showed substantial reductions in MSE compared with baseline methods. This indicates that these variables contain relationships with other attributes that can be effectively exploited by regression- and distance-based techniques.

c. **Some variables are weakly predictable from the available features.**

For gender and hypertension, the Mean baseline achieved the lowest error. This suggests that the remaining attributes provide limited predictive information for these variables, making more complex models unnecessary.

7. Recommendations

a. **Use the selected per-column imputation strategy in practice.**

The evaluation shows that different columns benefit from different techniques. Therefore, missing values should be filled using the best-performing method for each column rather than applying a single imputation technique to the entire dataset.

b. **Keep the held-out, multi-split selection.**

Averaging MSE across multiple 80/20 train-test splits provides a more stable estimate of model performance, particularly for small datasets. Future studies may increase the number of splits to obtain even more reliable estimates.

8. Limitations

a. MSE is scale-dependent and should only be compared within each feature.
b. Alphabetical ordinal encoding introduces arbitrary ordering.
c. Computational cost increases due to multiple model training per feature.